BRAC UNIVERSITY

CSE461: Introduction to Robotics

Lab Project Report

Title: Autonomous Heat Stroke Detection & Response Robot

by

[Group : 05]
[Section : 2]

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Abstract

Heat stroke is a serious health problem that happens when the body gets too hot due to high temperature and humidity. People who work outside, athletes, or elderly people are often at risk. The main goal of our project is to build a robot that can automatically detect the risk of heat stroke and take quick actions to cool the person down.

The robot uses several sensors. The MLX90614 sensor measures human body temperature, the DHT11 checks the surrounding temperature and humidity, and the MPU6050 detects if someone has fainted or stayed still for a long time. All the data is processed by the ESP32 microcontroller. When a risk is detected, the robot turns on a fan, sprays water mist, and uses LED lights for alerts. If needed, it can also send a notification through Wi-Fi to nearby people or authorities.

From our testing, the system worked well and only used its cooling functions when there was real danger, which saves power. In the future, this robot could be used in places like factories, construction sites, schools, or elderly care centers. This shows how a simple and low-cost robot can help protect people's health and even save lives in hot weather.

Keywords

Heat stroke, ESP32 microcontroller, Sensors, Automation, Cooling system

1. Introduction

Heat stroke is a dangerous condition that happens when the human body cannot control its temperature in hot and humid environments. In countries like Bangladesh, this problem is very common, especially for outdoor workers, athletes, and elderly people. If it is not treated quickly, heat stroke can even become life-threatening. Because of this, there is a strong need for systems that can detect early signs of heat stroke and give fast support.

The main objective of our project is to design and build an autonomous robot that can monitor the human body temperature, the surrounding environment, and movement patterns to identify the risk of heat stroke. Once the robot detects any risky situation, it will automatically respond by turning on a fan, spraying water, and giving alerts. It can also send notifications through Wi-Fi so that nearby people or authorities can take further action.

This project is closely related to the course CSE360: Computer Interfacing, because the system combines sensors, actuators, and a microcontroller (ESP32) to work together. The microcontroller reads sensor data, processes it, and controls the output devices in real time. Through this project, we learned how to connect hardware components, use digital communication protocols, and implement control logic in practical applications.

2. Related Work/ Inspiration

There have been different types of systems developed before to help reduce the risk of heat stroke. Some researchers worked on wearable devices that people can carry with them. For example, IoT-based wearables have been made that measure body temperature, heart rate, and humidity, then send alerts through a mobile app if the person is at risk.[1] Other projects used non-contact sensors to measure deep body temperature and warn users in advance. In addition, some researchers even tested drone-based systems that can fly over hot areas, detect heat stress with cameras, and spray water to cool down workers.[2]

These works inspired our idea, but our project is different in several ways. Most of the existing systems are either expensive or too complex for daily use. Wearable devices can only warn the user but don't actually provide cooling, while drone systems might help in some cases but aren't really practical for everyday environments. On the other hand, our robot is affordable, simple, and combines both detection and response. It not only monitors the body and environment but also takes direct action by running a fan, spraying water, and giving visual alerts.

With this approach, our system aims to bridge an important gap: it offers a compact and affordable robot that doesn't just issue warnings but also delivers immediate cooling support. This makes it highly practical for use in places like factories, schools, or households particularly in developing countries where resources are limited and heat stroke poses a serious health threat.

3. Technical Approach

System Architecture

Our robot has three main parts: sensors, a controller, and output devices.

- The **sensors** (MLX90614, DHT11, and MPU6050) check body temperature, room temperature & humidity, and movement.
- The **controller** (ESP32) collects all this data and decides what action to take.
- The **outputs** (fan, water pump, LED, Wi-Fi) are activated if the robot finds any danger signs.

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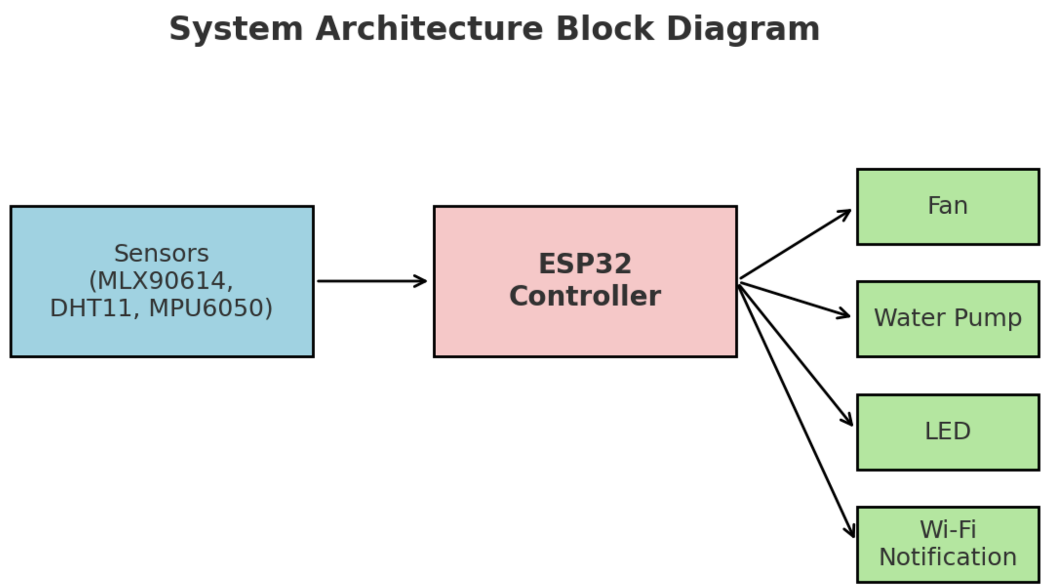
[1] **Tamura et al. (2022).** Tamura, T.; Lee, C. H.; et al. An Advanced Internet of Things System for Heatstroke Prevention: Design and Prototyping. *Sensors* **2022**, *22*(24), 9781016. https://doi.org/10.3390/s22249716 PMC

[2] **Hu & Assaad (2024).** Hu, X.; Assaad, R. H. Automated Heat Stress Monitoring and Water-Spraying Robotic System for Improving Work Conditions Using Drone (UAV) Infrared Thermography. In *Proceedings of the Construction Research Congress 2024*, 2024; pp. 1–10. https://doi.org/10.1061/9780784485262.076



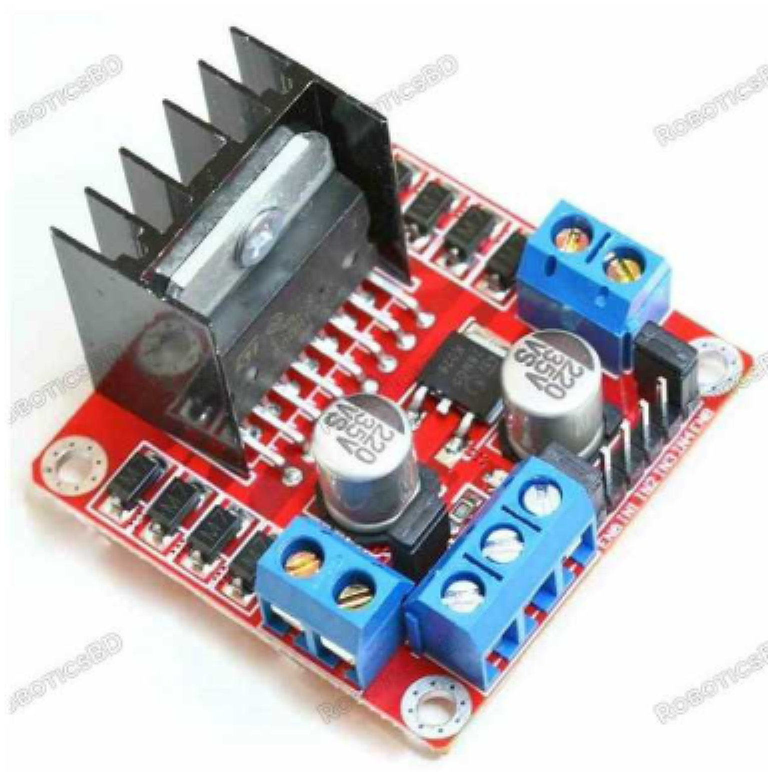
Workflow of the System

1) The robot starts and reads values from all sensors.
2) It compares the readings with set safety limits.
    a) If values are normal → it keeps monitoring.
    b) If high body or air temperature is detected → the fan turns on.
    c) If fainting or stillness is found → the pump sprays mist and the alert system activates.
    d) If the problem continues → a Wi-Fi message is sent.

Components Used:

Hardware:

1) L298N Motor Driver

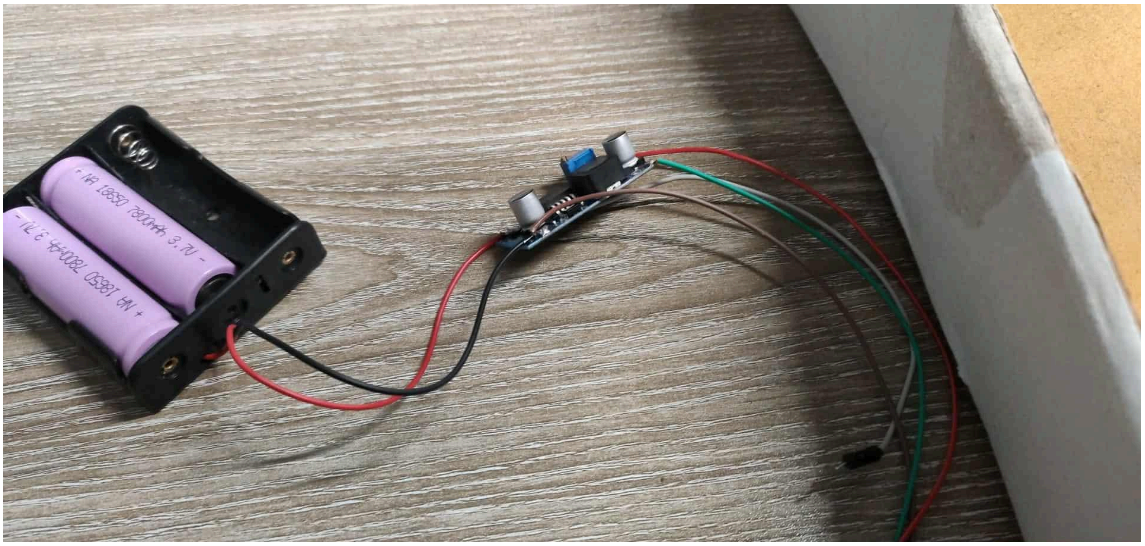


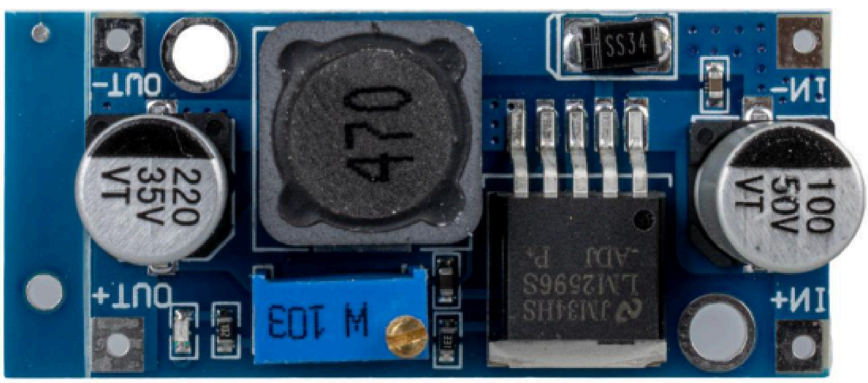
The L298N Motor Driver Module is a dual H-Bridge motor driver that can control the speed and direction of up to 2 DC motors (7V–35V, max 2A per channel) or one stepper motor. It features onboard 78M05 voltage regulator to supply 5V logic, with jumpers for enabling/disabling. Widely used in robotics projects, it allows easy interfacing with microcontrollers like ESP32 and Arduino.

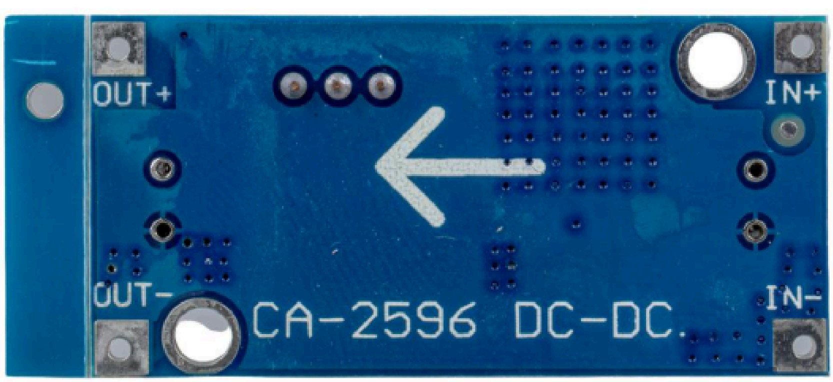
2. ESP32 Development Board

The ESP32 is a powerful Wi-Fi + Bluetooth-enabled microcontroller with a dual-core processor running at up to 240 MHz, and 520 KB SRAM. It operates on 3.0V–3.6V logic (commonly powered by 5V USB with onboard regulator). With built-in ADCs, PWMs, UART, SPI, and I2C interfaces, it is ideal for IoT, robotics, and sensor-based projects.

3. Battery Holder with CA2596 DC-DC Converter and 3× 18650 Li-Ion Batteries (3.7V, 7800mAh each)

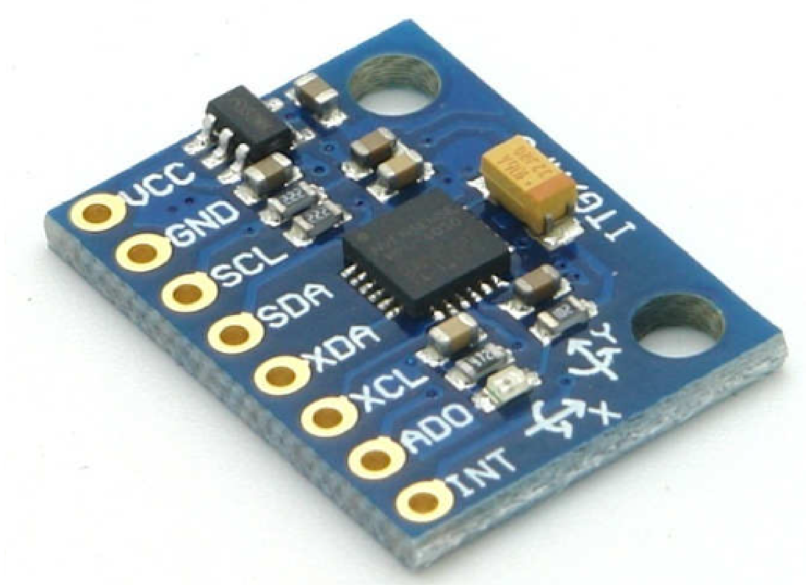






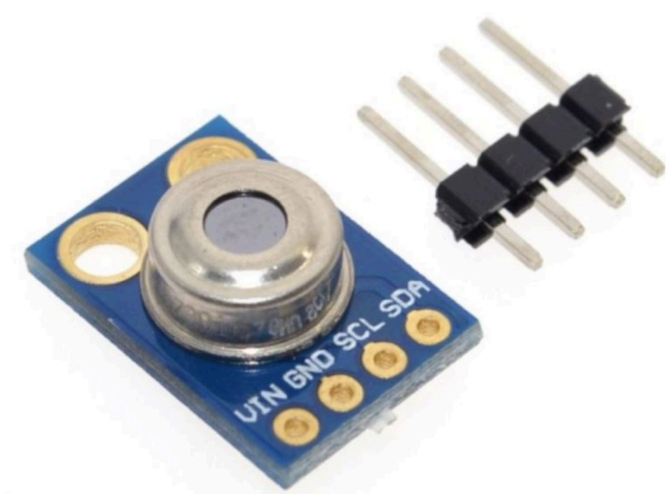
This battery pack combines two 18650 rechargeable Li-Ion cells (3.7V nominal, 4.2V fully charged, 7800mAh capacity each) inside a holder, paired with a CA2596 buck converter for stable voltage regulation (typically 1.25V–35V adjustable output, 2A–3A max - Adjusted to 5.33V). This setup ensures reliable and regulated power for ESP32, motors, and sensors.

4. MPU-6050 (Gyroscope + Accelerometer Module)



The MPU-6050 is a 6-axis MEMS sensor combining a 3-axis gyroscope and a 3-axis accelerometer. It communicates via I2C (3.3V/5V tolerant) and operates at 2.375V–3.46V. Commonly used for motion tracking, orientation detection, and stability control in robotics.

5. GY-906 (MLX90614 Infrared Temperature Sensor)



The GY-906 module uses the MLX90614 IR thermometer, capable of non-contact temperature measurements in the range of -70°C to +380°C with 0.5°C accuracy. It communicates via I2C (3.3V/5V) and operates at 3.6V–5V. Ideal for body temperature or object heat detection in health-monitoring and automation projects.

6. DHT11 Temperature and Humidity Sensor

The DHT11 is a basic, low-cost digital temperature (0–50°C, ±2°C accuracy) and humidity (20–90%, ±5% accuracy) sensor. It operates at 3.5V–5.5V and communicates via a single-wire digital interface. Suitable for indoor environmental monitoring and basic automation systems.

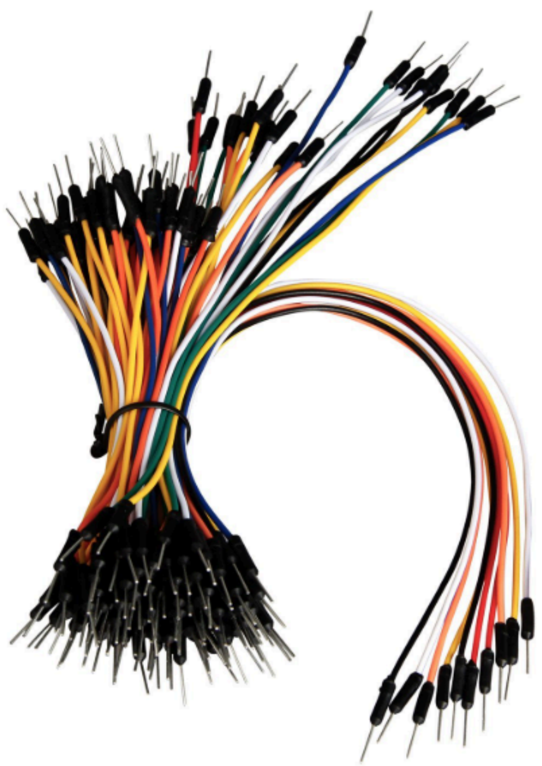
7. JD-4007S5L2 Brushless DC Fan (5V, 0.08A)

This is a compact 5V brushless DC cooling fan, rated at 0.08A (≈0.4W power consumption), commonly used for electronics cooling and thermal management. With low noise and reliable operation, it is ideal for providing airflow in small robotic or IoT devices where overheating is a concern.

8. TG-008 Li-Ion Battery Charger

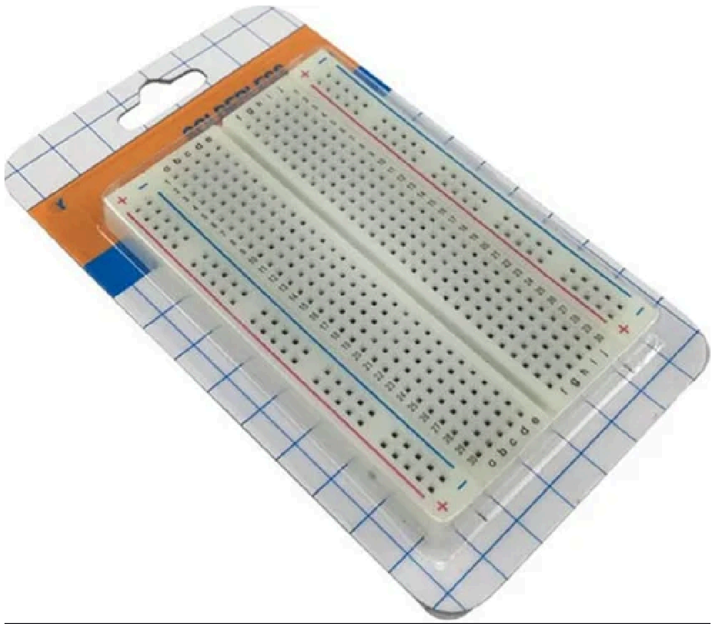
The TG-008 is a compact charger designed for 18650 and other Li-Ion batteries. It supports AC 110–240V input and provides 4.2V regulated charging output, ensuring safe and efficient charging with overcharge protection. Commonly used for portable robotics and IoT projects that rely on 18650 cells.

9. Jumper Wires



Jumper wires are flexible electrical connection cables used for prototyping and circuit testing. We used male-to-male, male-to-female, and female-to-female types. They carry low DC voltages (3.3V–5V) and small currents, making them ideal for connecting microcontrollers, sensors, and modules without soldering. Essential for building and testing circuits quickly.

10. Breadboard



A breadboard is a reusable solderless prototyping board that allows easy circuit construction without permanent connections. Standard boards operate up to 5V–12V DC and can handle around 1A per rail. Internally, rows are connected to allow quick placement of components like resistors, ICs, sensors, and jumper wires. Commonly used in learning, testing, and debugging circuits before final PCB design.

Software Tools:

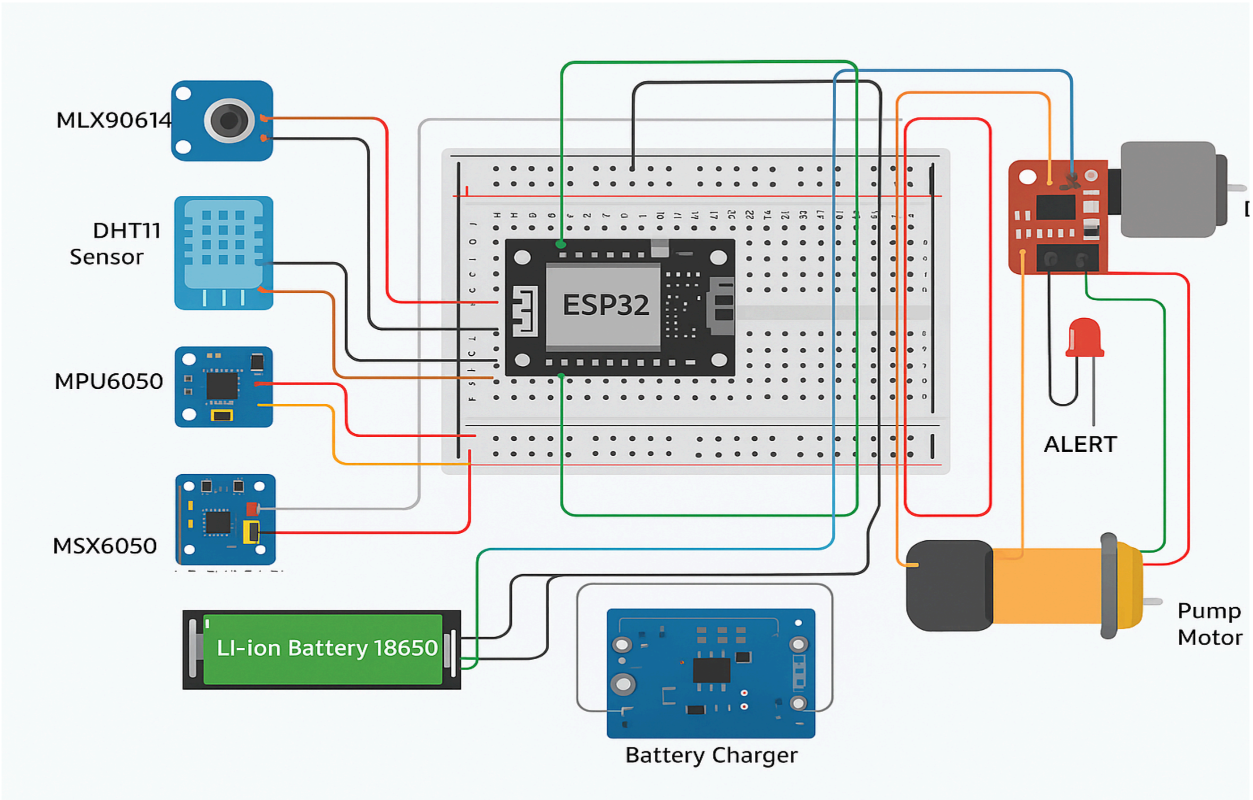
- **Arduino IDE** for writing, compiling, and uploading the ESP32 code.
- **ESP32 Board Package** installed via Arduino Boards Manager to program and configure the ESP32 Dev Module.
- **Arduino libraries** for sensors (Adafruit_MLX90614, DHT, Adafruit_MPU6050).
- **Serial Monitor** – debugging outputs.

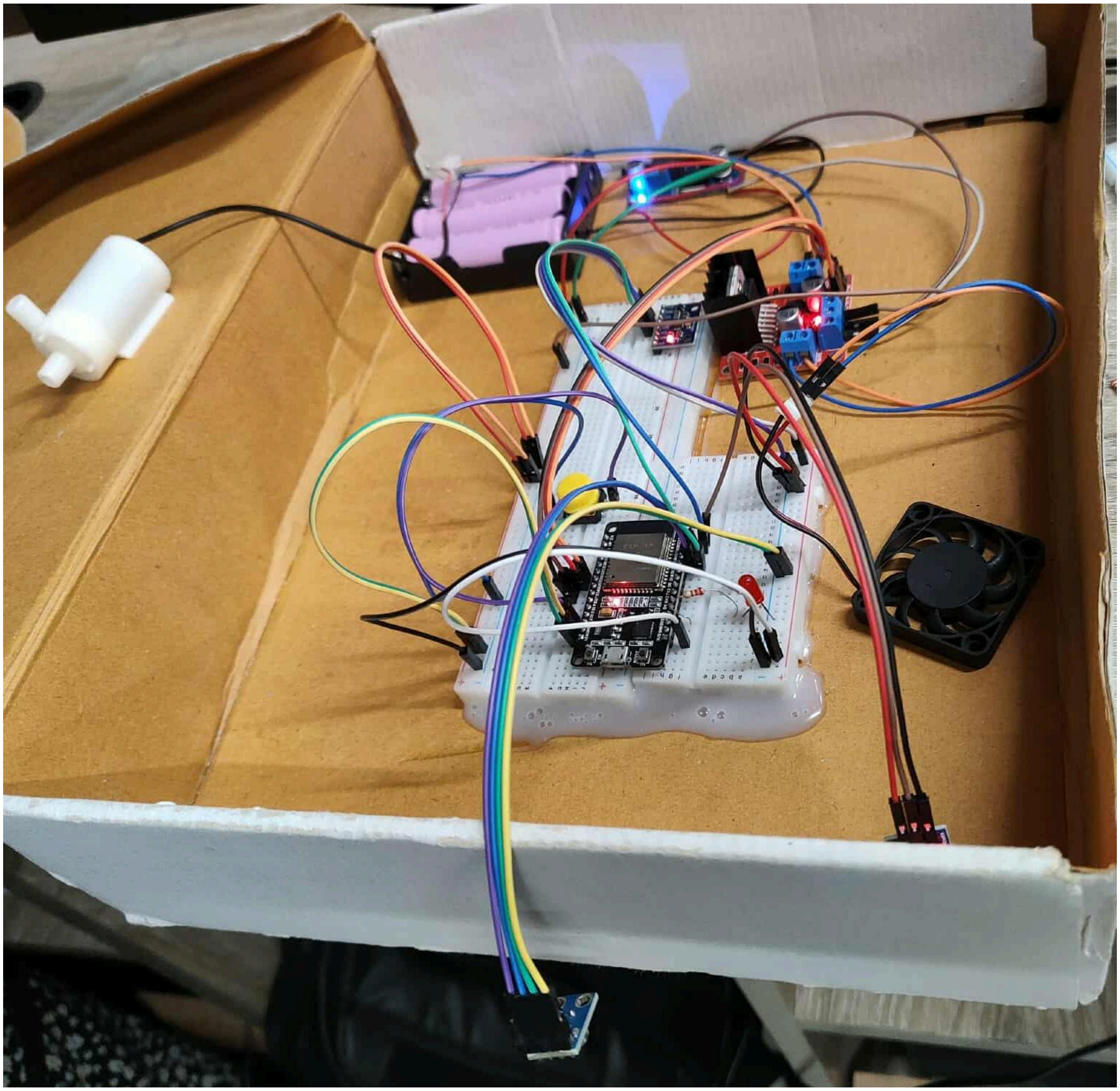
Cost Breakdown:

No	Components	Quantity	Unit Cost (BDT)	Total Cost (BDT)
1	ESP32 30P	1	550	550
2	MLX90614	1	1000	1000

3	DHT11 Sensor	1	200	200
4	MPU6050 Sensor	1	690	690
5	Mini Cooling Fan	1	190	190
6	LED	1	5	5
7	Battery and charger	1	500	500
8	Breadboard	2	150	300
9	DC water pump motor	1	450	450
10	Push Button	1	10	10
11	Motor Driver (L298N)	1	220	220
12	M-M, F-M,F-F Jumper Wires	3	150	450
13	Buck Converter	1	195	195
14	220ohm resistor	1	5	5
Total Cost (BDT)				4765

Circuit/Schematic:





Functionality:

1. Environment Monitoring (DHT11)

The DHT11 sensor is connected with Vcc to 3.3V, Data to GPIO 23, and GND to ground. It continuously reads the surrounding temperature and humidity levels. If the temperature rises above 35°C combined with high humidity, the system flags this as a risk condition. At that point, the cooling fan is activated to reduce heat stress in the environment. This ensures that the fan only runs when truly necessary, making the system energy efficient and responsive.

2. Body Temperature Monitoring (MLX90614)

The MLX90614 infrared sensor is connected with Vcc to 3.3V, SCL to GPIO 22, SDA to GPIO 21, and GND to ground. It measures human body temperature without any physical contact. If the measured body temperature goes above 40°C while the environment is already hot, the system interprets this as a possible heat stroke condition. In response, the ESP32 immediately turns on the fan for cooling and also sends an alert through WiFi, ensuring timely support.

3. Motion Monitoring (MPU6050)

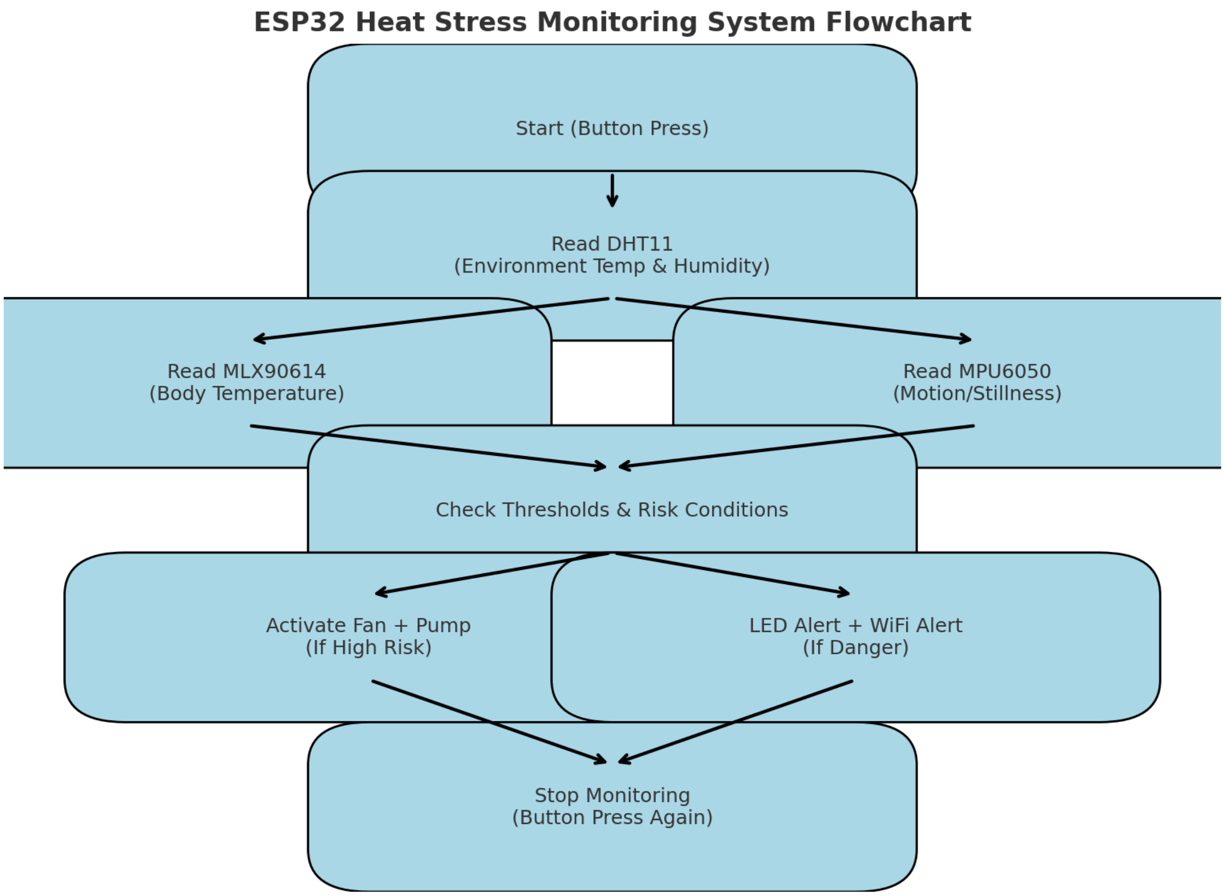
The MPU6050 accelerometer and gyroscope sensor is connected with Vcc to 3.3V, SCL to GPIO 22, SDA to GPIO 21, and GND to ground. It is responsible for monitoring motion and movement patterns. If the system detects prolonged stillness, it flags this as a danger event that may indicate unconsciousness due to heat stress. In such cases, the ESP32 sends an alert through WiFi and also activates the LED for visible warning. If regular motion is detected, then the system continues normal temperature sensing without raising alerts.

4. Actuation & Power Management

The fan for cooling airflow is connected to the motor driver, with OUT1 wired to fan Vcc and OUT2 to fan GND, while the control inputs IN1 and IN2 are connected to GPIO 32 and GPIO 33 respectively. The water pump for mist spraying is also connected through the motor driver, with OUT3 wired to pump GND and OUT4 to pump Vcc, while the control inputs IN3 and IN4 are connected to GPIO 25 and GPIO 26. Additionally, an LED is connected to GPIO 2 to serve as a visual alert indicator. When risk is detected from the DHT11, MLX90614, or MPU6050 sensors, the ESP32 activates the fan and pump together for active cooling, switches on the LED for visible alerts, and sends an alert message over WiFi.

5. System Control (Push Button , WiFi Alerts)

A push button is used to control the entire system, with one pin connected to ground and the opposite diagonal pin connected to GPIO 12. Pressing the button starts the monitoring process, activating all sensors, while pressing it again stops the monitoring. This gives manual control over when the system should operate. For communication, the ESP32 uses its built-in WiFi capability through the WiFi.h library. When any danger or risk is flagged such as heat stroke risk, prolonged stillness, or extreme environmental conditions the ESP32 sends alert messages with sensor data via WiFi, allowing nearby people or a monitoring system to be informed immediately.



Challenges and Solutions

The problems we faced in our project and the ways we solved them are explained below:

Sensor Accuracy Problem

The DHT11 sensor was not very precise when measuring temperature and humidity. Sometimes the readings went up and down even in the same condition. To solve this, we did not rely on a single reading. Instead, we took multiple readings and then calculated the average. This made the results more stable and trustworthy.

Motor Power Issue

The fan motor and pump motor needed more current than the ESP32 could provide directly. At first, the motors were not running properly. We solved this by using the L298N motor driver. The driver allowed us to control the motors with signals from the ESP32, while the actual power came from a separate battery supply.

Noise in MPU6050 Data

The MPU6050 sensor gave some random noise in the motion readings, which sometimes created false alarms. To handle this, we used a threshold system. That means the ESP32 only considered it as no-movement or fainting if the sensor showed the same condition for a certain time. We also added a small delay in the code so that sudden random changes were ignored.

Water Splash Risk

Because our robot sprays mist, there was a chance that water could damage the electronic parts like ESP32 or the motor driver. To protect them, we used a simple casing around the circuits. We also placed the electronics in a position where the mist could not directly reach them. This reduced the risk of short circuits or damage.

Testing Limitation

We could not test the system on real patients because it is risky. This was a big limitation. To still check the system, we created fake conditions. For example, we used hot water bottles to simulate high body temperature and heaters to simulate a hot environment. This helped us to test whether the sensors and actuators responded correctly in those situations.

4. Sustainability & Impact

Sustainability

Our robot is designed to be energy efficient. The sensors do not run continuously but take readings at short intervals, which saves power. The actuators (fan and pump) are only turned on when the system detects risk, so no extra energy is wasted. Rechargeable Li-ion batteries are used, which reduces the need for disposable batteries and makes the system more environmentally friendly. Also, many of the parts like the ESP32, sensors, and motors can be reused in future projects, which supports recycling and reduces e-waste.

Impact

This project has a strong social and community benefit. In hot countries like Bangladesh, heat stroke is a big problem for workers, elderly people, and students. Our robot can give quick first aid support by cooling the body and sending alerts before medical help arrives. In factories, construction sites, or farms, this can protect workers and reduce accidents. In schools or elderly homes, it can provide safety and peace of mind. On a larger scale, using such robots can improve occupational health and safety standards in industries.

Future Work

Scopes

Upgrading the design could further increase impact in real life implementation.

Advanced Sensing: Integrate thermal cameras, LiDAR, or wearable sign sensors for more personalized risk detection. Integrate more sophisticated sensors, such as pulse rate and oxygen level monitors, to enhance health tracking capabilities.

Smart Mobility: Enhance the robot's mobility by equipping it with wheels, allowing it to navigate and reach people autonomously.

Enhanced Cooling: Add misting systems or deploy lightweight canopies for passive shade.

Edge Computing & Connectivity: Use Jetson embedded GPUs for real-time thermal mapping and integrate LTE/5G, mesh, or satellite comms for remote operation.

Human Interaction: Add touch screens, voice prompts, or mobile apps for richer interaction and notifications. Enable connectivity with mobile applications or hospital networks to ensure faster emergency alerts and responses.

Sustainability : Implement solar panels for charging, making the system more energy-efficient and environmentally friendly.

Scalability options

In order to enhance it further some measures can be taken:

1. **Privacy**: Avoid capturing identifiable images, use thermal-only imaging, anonymizing data through on-board processing.
2. **Medical Disclaimer**: Emphasize that the robot detects suspected heat stress, not diagnosing; human intervention is needed for validation and treatment.
3. **Fail-Safes**: Incorporate redundant power, door-lock overrides, and emergency stop systems.
4. **Regulation and Liability**: Position the robot as an assistive device, not a medical tool, and engage institutional review processes where applicable.
5. **Hygiene**: If implementing misting, ensure UV filtration or replaceable nozzles for sanitary use.

Limitations

Even though the project was successful, we faced some limitations:

01. **Hardware:** The DHT11 sensor is less accurate compared to advanced sensors. The pump motor is also small and cannot spray water for a long time.

02. **Software:** We used simple threshold logic, a more advanced algorithm like machine learning could give better predictions.

03. **Power:** The system depends fully on batteries, so it cannot run for very long without recharging.

04. **Testing:** We could not test on real patients, so real-world effectiveness is still unproven.

5. Results & Discussion

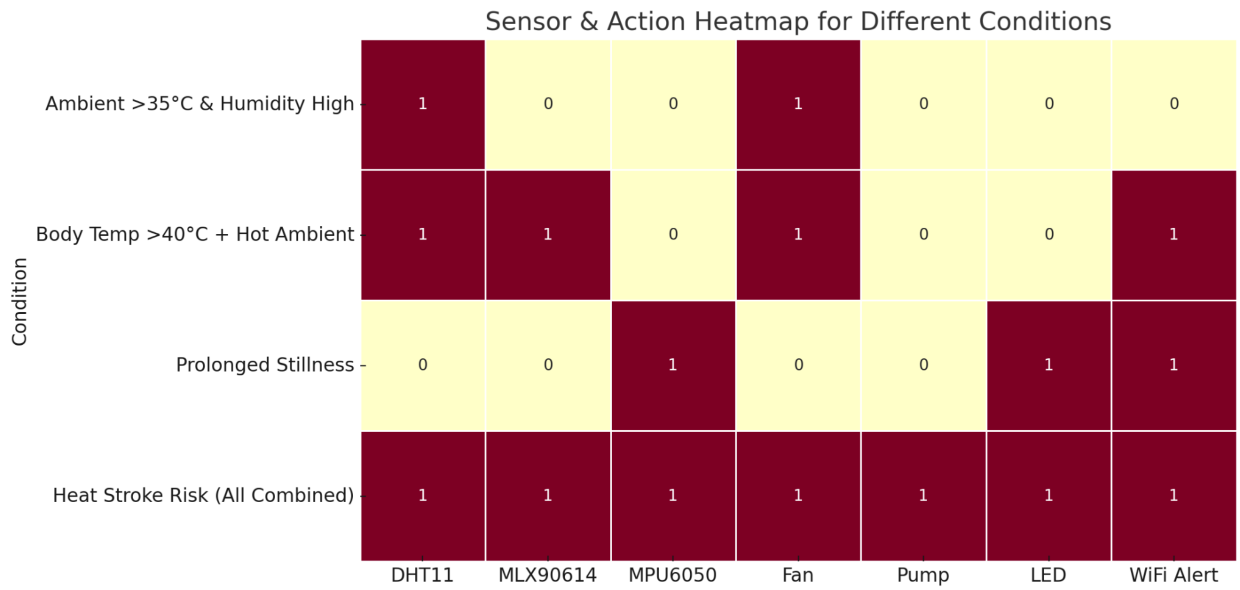
We tested the robot by creating fake hot and fainting conditions:

Test: A hot water bottle was used to simulate a high body temperature and a lighter was used to raise air temperature.The MPU6050 was kept flat and motionless.

Findings table:

Condition	Sensor triggered	Action Taken	Alert sent
Ambient > 35°C & Humidity High	DHT11	Fan ON	No
Body Temp > 40°C + Hot Ambient	MLX90614 , DHT11	Fan ON, WiFi Alert	Yes
Prolonged Stillness	MPU6050	LED ON, WiFi Alert	Yes
Heat Stroke Risk (all combined)	All Sensors	Fan and Pump ON, LED ON, WiFi Alert	Yes

Visual heatmap chart of which sensors activates:



Discussion

From the test results, we can say that the robot worked properly according to its goals. The sensors were able to measure both the surrounding temperature and the body temperature correctly. When the temperature was normal, the fan stayed off which helped to save power. But as soon as the temperature crossed the threshold level, the fan turned on automatically and controlled the heat. For movement detection, the system also worked well. In case of fainting or unusual stillness, the pump motor got activated, which showed that the robot could detect abnormal conditions. Along with that, the LED light and Wi-Fi notification provided extra safety by giving alerts to others. Overall, the tests proved that each part of the robot functioned as expected.

The results also show that the system avoids false alarms most of the time, because of the threshold-based logic. However, in some cases small fluctuations in sensor readings caused the fan to turn on/off quickly. This was reduced by averaging multiple readings.

Overall, the project proves that a simple microcontroller-based system can detect early signs of heat stroke and take quick action. It met the main objectives of monitoring, cooling, and alerting.

6. Conclusion

The main purpose of this project was to build a robot that can detect the risk of heat stroke and take quick action to help the person. We used different sensors to measure body temperature, environment conditions, and movement, and connected them to an ESP32 microcontroller. Based on the readings, the robot could turn on a fan, spray water, and send alerts to people nearby.

From our testing, the system worked as expected. It only activated when risk conditions were detected, which helped to save energy. The fan and pump responded correctly to high temperature and stillness, and the LED alerts gave extra safety. Wi-Fi notifications added another level of support, which makes the system more practical in real life.

This project also shows the importance of sustainability and impact. We used rechargeable batteries, reusable components, and an energy-efficient design, which makes the robot environmentally friendly. At the same time, the system has a strong social impact because it can be used in factories, schools, sports events, or elderly care centers to protect people from heat stroke.

In conclusion, our project successfully met its objectives and showed that even a small, low-cost robot can make a real difference in health and safety. With future improvements, it can become even more reliable and widely used in the community.

7. Contribution

Name	ID	Role(s) / Responsibilities	Specific Contributions
Maisha Farzana Joyee	21201065	Hardware design, Circuit building	• Worked on hardware design and diagrams for the implementation • Integrated the sensors : MLX90614, DHT11, MPU9060 with ESP32 for reliable temperature, humidity, body temperature, and motion monitoring. • Wired the cooling fan, water pump, and LED alert through the motor driver and ensured correct GPIO control. • Implemented push button functionality to start and stop the monitoring system manually. • Analysed results from the system
Sohanur Rahman Khan	24241335	Circuit design support, Arduino Coding	• Implemented WiFi alert notifications. • Ensured proper power(5v) supply from motor driver for microcontroller ESP32 • Wrote parts of the ESP32 code for reading DHT11 or MPU6050 sensor values. • Helped implement threshold logic for triggering the fan.

			- Tested the individual components through test codes - Contributed to create flowchart for the system workflow.
Abdun Noor Adnan	21301707	Arduino Coding support, Testing & Troubleshooting	- Ran tests with a hot water bottle and heater to simulate high temperatures. - Checked sensor readings and noted errors during testing. - Ensured efficient operation of sensors and actuators, activating cooling only when needed to save energy.
Tahiya Haider	21201023	Hardware setup, Documentation & Report Writing	- Prepared parts of the report (e.g., Technical Approach or Results & Discussion). - Contributed to Created block diagram & flowchart for the system workflow. - Worked in setting up sensors, actuators with ESP32
Aovejeet Saha	21301259	Power Management Setup & Report diagrams	- Helped connect Li-ion batteries and charger. - Ensured the ESP32 and motors had stable power supply. - Took part in testing by simulating high body temperature and checking sensor accuracy. - Contributed to report writing by preparing the Technical Approach section and drawing the block diagram. - Suggested improvements for future work (adding new sensors, mobility, solar charging).

References

1.Hu, X., & Assaad, R. H. (2024). Automated heat stress monitoring and water-spraying robotic system for improving work conditions using drone (UAV) infrared thermography. *Proceedings of the Construction Research Congress 2024*, 1–10. https://doi.org/10.1061/9780784485262.076

2.Silawarawet, K., Kaewchukul, P., & Saadprai, S. (2025). Heat Stroke Warning System Prototype for Athletes: A Pilot Study. *Sensors*, *25*(2), 294. https://doi.org/10.3390/s25020294

3.Tamura, T., Lee, C. H., et al. (2022). An advanced Internet of Things system for heatstroke prevention: Design and prototyping. *Sensors, 22*(24), 9781016. https://doi.org/10.3390/s22249716

4.EnVisio. (2025, March 7). *How to Adjust Voltage on an LM2596 in Seconds!* [Video]. YouTube. https://www.youtube.com/watch?v=dFXKUqf0dwM&t=42s